

Exercise 2: E-commerce Platform Search Function

Code:

```
using System;
```

```
class Product
{
    public int ProductId;
    public string ProductName;
    public string Category;

    public Product(int id, string name, string category)
    {
        ProductId = id;
        ProductName = name;
        Category = category;
    }
}
```

```
class Program
{
    // Linear Search
    static Product LinearSearch(Product[] products, int id)
    {
        foreach (Product p in products)
        {
            if (p.ProductId == id)
                return p;
        }

        return null;
    }

    // Binary Search
    static Product BinarySearch(Product[] products, int id)
    {
        int low = 0;
        int high = products.Length - 1;

        while (low <= high)
        {
            int mid = (low + high) / 2;
```

```

        if (products[mid].ProductId == id)
            return products[mid];

        if (products[mid].ProductId < id)
            low = mid + 1;
        else
            high = mid - 1;
    }

    return null;
}

static void Main()
{
    Product[] products =
    {
        new Product(101, "Laptop", "Electronics"),
        new Product(102, "Phone", "Electronics"),
        new Product(103, "Shoes", "Fashion"),
        new Product(104, "Watch", "Accessories"),
        new Product(105, "Book", "Education")
    };

    Console.WriteLine("Enter Product ID to search: ");
    int id = Convert.ToInt32(Console.ReadLine());

    Console.WriteLine("\nLinear Search");

    Product result1 = LinearSearch(products, id);

    if(result1 != null)
    {
        Console.WriteLine("Found");
        Console.WriteLine("ID : " + result1.ProductId);
        Console.WriteLine("Name : " + result1.ProductName);
        Console.WriteLine("Category : " + result1.Category);
    }
    else
    {
        Console.WriteLine("Product Not Found");
    }

    Console.WriteLine("\nBinary Search");

```

```

Product result2 = BinarySearch(products, id);

if(result2 != null)
{
    Console.WriteLine("Found");
    Console.WriteLine("ID : " + result2.ProductId);
    Console.WriteLine("Name : " + result2.ProductName);
    Console.WriteLine("Category : " + result2.Category);
}
else
{
    Console.WriteLine("Product Not Found");
}

Console.WriteLine("\nTime Complexity Comparison");
Console.WriteLine("Linear Search : O(n)");
Console.WriteLine("Binary Search : O(log n)");
}
}

```

Output:

```

C:\Users\suman\OneDrive\Desktop\CTS\SingletonPattern\Program
Enter Product ID to search: 103

Linear Search
Found
ID : 103
Name : Shoes
Category : Fashion

Binary Search
Found
ID : 103
Name : Shoes
Category : Fashion

Time Complexity Comparison
Linear Search : O(n)
Binary Search : O(log n)

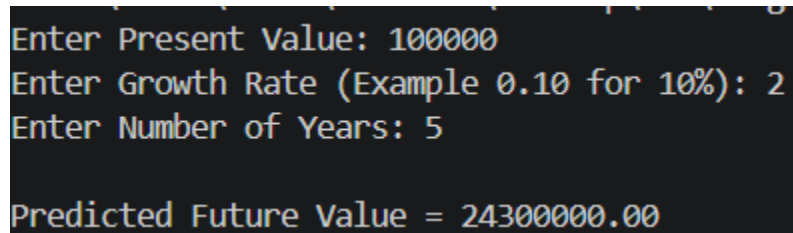
```

Exercise 7: Financial Forecasting

Code:

```
using System;
class FinancialForecast
{
    static double PredictFutureValue(double presentValue, double growthRate, int years)
    {
        if (years == 0)
            return presentValue;
        return PredictFutureValue(presentValue, growthRate, years - 1) * (1 + growthRate);
    }
    static void Main()
    {
        Console.Write("Enter Present Value: ");
        double present = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter Growth Rate (Example 0.10 for 10%): ");
        double rate = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter Number of Years: ");
        int years = Convert.ToInt32(Console.ReadLine());
        double future = PredictFutureValue(present, rate, years);
        Console.WriteLine("\nPredicted Future Value = " + future.ToString("F2"));
    }
}
```

Output:



```
Enter Present Value: 100000
Enter Growth Rate (Example 0.10 for 10%): 2
Enter Number of Years: 5

Predicted Future Value = 24300000.00
```